Tharun Kumar Kumaravel

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SKILLS

Languages: Python, C, C++, Java
Web: HTML, CSS, JavaScript, Node.js, React
AI/ML: Machine Learning, Deep Learning, Computer Vision, Transfer Learning, Ensemble Learning
Py-Libraries: PyTorch, NumPy, Pandas, Streamlit
Blockchain: Solidity, Ethereum, Smart Contracts, ERC-721/NFTs, Ethers.js
Tools: Git, GitHub, JPackage, Docker
Soft Skills: Problem Solving, Adaptability, Analytical Thinking

PROJECTS

Blockchain Identity & Asset Management Platform | GitHub — Sep' 26

- Created a blockchain-based platform prototype for real-world identity, access-control, and asset-ownership management using smart contracts and NFTs.
- Engineered a security-first architecture with RBAC (role-based access control), multi-signature approvals, an emergency pause feature, and automated anomaly detection to prevent misuse of admin privileges.
- Developed a full-stack solution (React frontend + Node.js backend) including a public, wallet-free QR verification page for real-time asset tracking.
  Tech Stack: Blockchain, React.js, JavaScript, Node.js, Express.js, Solidity, Hardhat, ethers.js, OpenZeppelin, SQLite, Docker

Pyllon - Plant Disease Detection AI | GitHub — May' 26

- Built an AI-powered web application for multiclass plant disease detection with dynamic per-plant model loading and AI-assisted disease guidance through an API-powered chatbot.
- Trained plant-specific ensemble models using a three-phase transfer-learning pipeline with MixUp, label smoothing, EMA, and 5-pass TTA, achieving 90–99% validation accuracy across the models.
- Optimized cloud inference with dynamic quantization, reducing each plant-specific model pair's size from ~198 MB to ~50 MB while operating within Streamlit's memory constraints.
  Tech Stack: PyTorch, Python, Streamlit, OpenCV, Transfer learning, Deep learning, Ensemble learning, ConvNeXt-Tiny, EfficientNetV2-S

RTOS ThreadVision - OS Thread Simulator | GitHub — Apr' 26

- Constructed a zero-dependency multicore CPU scheduling and multithreading simulator in Java Swing for interactive visualization of thread execution and core-level scheduling.
- Implemented thread synchronization and real-time visualization through interactive Gantt charts, per-core execution views, and thread-state monitoring.
- Packaged the simulator as a standalone Windows application using jpackage and open-sourced it for community learning and contribution.
  Tech Stack: Java, Java Swing, JPackage

CERTIFICATES

Data Structures and Algorithms – Summer Training | Board Infinity — Jul' 26
Programming in Java | iamNeo — May' 26
Programming Using C++ | Infosys Springboard — Aug' 25
Introduction to DSA | Coding Tantra — Jan' 25

EDUCATION

Lovely Professional University — Phagwara, Punjab
Bachelor of Technology – Computer Science and Engineering — Aug' 24 – Present
CGPA: 7.83

Thamarai International School — Thanjavur, Tamil Nadu
Senior Secondary (Intermediate) – MBPC — Mar' 23 – Mar' 24
Percentage: 92%